SDN Learning Switch Controller Report

Name : Poojitha Lakshmi Bhaskara Raghavendra

SRN : PES1UG24AM384

SEC : G

1. Introduction

This project implements a Software Defined Networking (SDN) based learning switch using the POX controller and Mininet. The objective is to dynamically learn MAC addresses and install forwarding rules to enable efficient packet transmission.

2. Objectives

- Understand SDN architecture
- Implement MAC learning logic
- Install dynamic flow rules
- Validate network behavior

3. Tools Used

Mininet, POX Controller, Open vSwitch, Ubuntu

4. Methodology

The controller listens for PacketIn events from switches. It learns source MAC addresses and maps them to ports. If the destination MAC is known, a flow rule is installed. Otherwise, the packet is flooded.

5. Implementation

A custom POX controller (learning_switch.py) was implemented. The controller handles PacketIn events and installs flow rules dynamically.

POX Setup:

```
poojitha@poojitha:~/pox$ cd ~/pox
poojitha@poojitha:~/pox$ ./pox.py log.level --DEBUG forwarding.l2_learning
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.12.3/Mar 3 2026 12:15:18)
DEBUG:core:Platform is Linux-6.17.0-20-generic-x86_64-with-glibc2.39
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
```

6. Results

- Successful communication between hosts (0% packet loss)
- Flow rules installed in switch
- Efficient forwarding after learning

7. Test Scenarios

1. Normal Scenario: pingall successful

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

```
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.369 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.252 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.144 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.112 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.148 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4114ms
rtt min/avg/max/mdev = 0.112/0.205/0.369/0.094 ms
mininet> h2 ping -c 5 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=3.41 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.943 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.061 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.082 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.199 ms

--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4146ms
rtt min/avg/max/mdev = 0.061/0.938/3.409/1.277 ms
mininet> h2 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.089 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.092 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2106ms
rtt min/avg/max/mdev = 0.036/0.072/0.092/0.025 ms
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.23 ms
```

```
--- 10.0.0.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4146ms
rtt min/avg/max/mdev = 0.061/0.938/3.409/1.277 ms
mininet> h2 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.089 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.092 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2106ms
rtt min/avg/max/mdev = 0.036/0.072/0.092/0.025 ms
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.23 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.466 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.076 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2038ms
rtt min/avg/max/mdev = 0.076/1.258/3.232/1.404 ms
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.08 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.551 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.112 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2048ms
rtt min/avg/max/mdev = 0.112/0.915/2.084/0.845 ms
mininet>
```

2. Failure Scenario: link down causes failure, link up restores connectivity

```
mininet> link s1 h2 down
mininet>
```

```
mininet> link s1 h2 down
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2036ms

mininet>
```

Failure (100% loss shown)

```
mininet> link s1 h2 up
mininet>
```

```
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.23 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.466 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.076 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2038ms
rtt min/avg/max/mdev = 0.076/1.258/3.232/1.404 ms
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.08 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.551 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.112 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2048ms
rtt min/avg/max/mdev = 0.112/0.915/2.084/0.845 ms
mininet>
```

```
mininet> link s1 h2 up
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.311 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.174 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2061ms
rtt min/avg/max/mdev = 0.174/3.673/10.535/4.852 ms
mininet>
```

Success(0% loss shown)

```
poojitha@poojitha:~$ sudo mn --controller=remote --topo=single,3 --switch=ovsk,protocols=OpenFlow10
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

3. OpenFlow dump

```
poojitha@poojitha:~$ sudo ovs-ofctl -O OpenFlow10 dump-flows s1
poojitha@poojitha:~$ sudo ovs-ofctl -O OpenFlow10 dump-flows s1
 cookie=0x0, duration=11.527s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=92:18:6e:9c:ca:a5,dl_ds
t=f6:44:8b:4d:18:ac,nw_src=10.0.0.2,nw_dst=10.0.0.3,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth3"
 cookie=0x0, duration=11.526s, table=0, n_packets=5, n_bytes=490, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=f6:44:8b:4d:18:ac,dl_ds
t=92:18:6e:9c:ca:a5,nw_src=10.0.0.3,nw_dst=10.0.0.2,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth2"
 cookie=0x0, duration=5.958s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth3",vlan_tci=0x0000,dl_src=f6:44:8b:4d:18:ac,dl_dst=9
2:18:6e:9c:ca:a5,arp_spa=10.0.0.3,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s1-eth2"
 cookie=0x0, duration=5.957s, table=0, n_packets=1, n_bytes=42, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth2",vlan_tci=0x0000,dl_src=92:18:6e:9c:ca:a5,dl_dst=f
6:44:8b:4d:18:ac,arp_spa=10.0.0.2,arp_tpa=10.0.0.3,arp_op=2 actions=output:"s1-eth3"
poojitha@poojitha:~$
```

8. Conclusion

The project successfully demonstrates SDN-based learning switch behavior using POX. It validates dynamic flow rule installation and efficient packet forwarding.